

UNIT–5: Computer Arithmetic & I/O Organization

1. **Explain addition and subtraction of signed numbers hookup.**
(RTU Exam: 2013, 2016, 2019)
2. **Explain Booth's multiplication algorithm with example.**
(RTU Exam: 2014, 2017, 2021, 2023)
3. **Explain restoring and non-restoring division algorithms.**
(RTU Exam: 2015, 2018, 2022)
4. **Explain floating-point arithmetic operations.**
(RTU Exam: 2016, 2019)
5. **Explain decimal arithmetic unit and BCD addition.**
(RTU Exam: 2014, 2018)
6. **Explain arithmetic logic unit (ALU) with diagram.**
(RTU Exam: 2013, 2017, 2020)
7. **Explain input-output organization of a computer system.**
(RTU Exam: 2015, 2019, 2022)
8. **Explain asynchronous data transfer techniques.**
(RTU Exam: 2016, 2020)
9. **Explain modes of data transfer (Programmed, Interrupt, DMA).**
(RTU Exam: 2014, 2018, 2021)
10. **Explain priority interrupt with daisy chain method.**
(RTU Exam: 2015, 2019)
11. **Explain DMA controller and its working.**
(RTU Exam: 2016, 2018, 2022)
12. **Explain IOP and CPU–IOP communication.**
(RTU Exam: 2017, 2021)